

ASAI Taj Delhi — Full Claude Briefing

Updated: Wednesday, 22 July 2026 · Project: **asai-taj-delhi** · Account: melvai@gmail.com · Repo: github.com/elonmuskisgod/asai-taj-delhi
Purpose: Single replication + handoff record for Claude (Firebase, firmware, WhatsApp, fleet concept). Supersedes partial 21 Jul work log for post-Twilio state.

Current status: One ESP32-S3 node publishes WiFi-direct to Taj RTDB every ~15s (live PUT + log POST). Live dashboard on Hosting. WhatsApp PM alerts via Twilio + Cloud Function onLiveWriteAlert (Blaze). Social Firebase untouched.

ASAI principle (non-negotiable): ASAI is about *all* data, not only alerts or aberrant readings. Never remove, thin, or skip history/logging to improve WiFi/cost/latency without explicit user approval. Prefer retry/queue/reconnect over dropping data.

1. URLs & accounts

What	Value
Live dashboard	https://asai-taj-delhi.web.app
Firebase console	https://console.firebase.google.com/project/asai-taj-delhi/overview
RTDB	https://asai-taj-delhi-default-rtdb.asia-southeast1.firebaseio.com
GitHub	https://github.com/elonmuskisgod/asai-taj-delhi
Local path	/Users/melroydmello/Documents/Arduino/asai-taj-delhi
Billing	Blaze (required for Cloud Functions → Twilio)
Do not use	Social / asaisocialproto2 / Social firmware

2. Repo layout

Path	Role
asai_dashboard/asai_dashboard.ino	Firmware v4.3 — sensors + OLED + WiFi HTTPS to Taj
asai_dashboard/secrets.h	WiFi SSID/password (gitignored)
public/index.html	Live single-node dashboard (Hosting)
functions/index.js	Cloud Function WhatsApp alerts (Twilio)
scripts/serial_to_taj.py	USB serial → RTDB fallback
scripts/whatsapp_watch.js	Local Twilio watcher (no Blaze needed)
scripts/seed_alerts_config.js	Seed PM phones + thresholds
scripts/deploy_whatsapp_function.sh	Secrets + firebase deploy --only functions
docs/fleet-intelligence-mockup.html	Static concept: 10-unit fleet intelligence
docs/TWILIO_WHATSAPP_ALERTS.md	Twilio sandbox setup
.cursor/rules/asai-data-first.mdc	Project rule: never compromise data

3. Hardware & data path

ESP32-S3 · BH1750 · AS7341 · BME680/BSEC · INMP441 · PMS7003 · OLED

1. Sensors sampled locally (~1s UI/Serial cadence)
2. Every ~15s: HTTPS PUT /ASAITajDelhi/live.json then POST /ASAITajDelhi/log.json (same payload)
3. Dashboard listens to live; timeline uses log
4. Cloud Function on live write → threshold check → Twilio WhatsApp

```
#define CLOUD_INTERVAL_MS 15000UL
// publishToTaj(): PUT live → delay 200ms → POST log (retry once on log fail)
// source field: "esp32_wifi"
// WiFi: DMello_4G (2.4 GHz only; ESP32 cannot use 5 GHz)
```

Orange flash in RTDB console every ~15s on live is expected. log should grow a new child each cycle (restored in v4.3 after a mistaken temporary disable — do not remove again).

4. RTDB paths

Path	Write	Purpose
/ASAITajDelhi/live	ESP PUT ~15s	Current snapshot
/ASAITajDelhi/log	ESP POST every sample	Full history (all data)
/ASAITajDelhi/delhi_outdoor	manual/seed	Outdoor US AQI compare
/ASAITajDelhi/alerts_config	seed / console	Thresholds + managers
/ASAITajDelhi/alerts_state	Cloud Function	Cooldownnotice / last-sent
/ASAITajDelhi/alerts_log	Cloud Function	WhatsApp send history

alerts_config (live RTDB)

```
{
  "enabled": true,
  "cooldown_ms": 10000,
  "thresholds": {
    "aqi": 200,
    "iaq": 200,
    "noise_level": 90,
    "temperature_high": 40,
    "temperature_low": 15
  },
  "managers": [
    { "name": "Melroy", "phone": "+919833020332", "enabled": true }
  ]
}
```

Cooldown 10s is for testing; pilot should use 1800000 (30 min).

Firebase console: after editing a value you must confirm with the grey control (not only click away) or the edit reverts.

5. Twilio WhatsApp (full story)

What was built

- Twilio account: Direct brand · With code · Notifications
- WhatsApp **Sandbox** sender typically whatsapp:+14155238886
- Secrets in Firebase Secret Manager: TWILIO_ACCOUNT_SID, TWILIO_AUTH_TOKEN, TWILIO_WHATSAPP_FROM
- Local secrets file: scripts/twilio.env (gitignored)
- Function: onLiveWriteAlert · region asia-southeast1 · instance asai-taj-delhi-default-rtdb

Why Blaze

Spark cannot make outbound HTTPS from Cloud Functions. Project upgraded to Blaze (pay-as-you-go; free quotas cover light use). Twilio WhatsApp is billed separately by Twilio (sandbox free for testing).

Sandbox constraints (critical)

- Recipient must WhatsApp join <code> to the sandbox number first
- Error **63015** = number not joined (or session expired ~3 days)
- Only joined numbers receive messages; Twilio branding on sandbox
- Production later: Meta/Twilio WhatsApp Business sender + templates

Proven end-to-end

Alert delivered and Twilio status read for noise threshold test. Example body:

```
ASAI Taj Delhi alert
Noise: 65 dBA (limit 40)
Temp 27.9°C · Lux 281.6667 · dB 65
21/7/2026, 7:08:26 pm
```

Deploy / seed commands

```
cd /Users/melroydmello/Documents/Arduino/asai-taj-delhi
./scripts/deploy_whatsapp_function.sh
PM_PHONE="+91XXXXXXXXXX" PM_NAME="Name" node scripts/seed_alerts_config.js
firebase functions:log --only onLiveWriteAlert
```

Alert debugging tips

- all hits in cooldown → wait 30 min or delete /ASAITajDelhi/alerts_state
- Arduino Serial can spike above threshold between publishes; Function only sees live.noise_level at ~15s write — hold loud sound across a publish
- Check Twilio message SID status if Firebase log shows queued but phone silent

6. Firmware evolution (code updates)

Version	Change
v3.x	Sensors + Serial + OLED only
v4.0	Firebase_ESP_Client → Taj RTDB
v4.1	HTTPS REST (lighter) PUT live
v4.2	WiFi auto-reconnect; live-only publish (log temporarily removed — mistake)

v4.3	live PUT + log POST every sample (current); log retry once
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Flash board: ESP32S3 Dev Module, CDC on boot, 8MB flash, default_8MB partition, OPI PSRAM. WiFi credentials in secrets.h.

```
FQBN example:
esp32:esp32:esp32s3:CDC0nBoot=cdc,USBMode=hwcdc,FlashSize=8M,PartitionScheme=default_8MB,PSRAM=opi
```

7. Dashboard behaviour

- Not a full page reload every 15s — Firebase `.on('value')` redraws when ESP writes
- “Live · Xs ago” ticks every 1s; outdoor AQI refresh every 15 min
- Stale detection if timestamp age > ~60s
- Single-node live UI; fleet concept is separate mockup

8. Fleet intelligence mockup (10 units)

Separate chat produced a static dead mockup imagining 10 coordinated ASAI nodes at Taj Delhi (cross-zone gradients, PM ingress, brand spectral signature, property rollups).

```
open docs/fleet-intelligence-mockup.html
# Not wired to Firebase — presentation concept only
```

9. Lessons learned

- ESP32 = 2.4 GHz WiFi only (DMello_5G fails; DMello_4G works)
- Do not use bare `millis()` as Firebase timestamp — dashboard marks years-stale
- Outdoor AQI: `aqi.in` shows US AQI; don’t mix CPCB math into that comparison
- Twilio sandbox 63015 = join required
- Never drop `/log` for WiFi stability without asking — ASAI is data-first
- Repo split: Taj lives in `asai-taj-delhi`, not `ac-ir-matrix`

10. Still open / Taj presentation

1. Production WhatsApp (leave sandbox) + approved templates
2. Lock RTDB security rules before wider sharing
3. Pilot multi-zone live fleet (mockup → real paths per node)
4. Pilot proposal one-pager + demo script
5. Commit uncommitted Twilio/functions/firmware v4.3 to GitHub if not yet pushed

11. How to brief Claude

1. Upload this PDF (or HTML) into a Claude Project
2. Add repo URL: <https://github.com/elonmuskisgod/asai-taj-delhi>
3. Optional attachments: `fleet-intelligence-mockup.html`, `TWILIO_WHATSAPP_ALERTS.md`
4. Do **not** upload `secrets.h` or `twilio.env`

